

ORIGINAL RESEARCH

A machine learning model for Alzheimer's disease prediction

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Abstract

Alzheimer's disease (AD) is a neurodegenerative disorder that mostly affects old aged people. Its symptoms are initially mild, but they get worse over time. Although this health disease has no cure, its early diagnosis can help to reduce its impacts. A methodology SMOTE-RF is proposed for AD prediction. Alzheimer's is predicted using machine learning algorithms. Performances of three algorithms decision tree, extreme gradient boosting (XGB), and random forest (RF) are evaluated in prediction. Open Access Series of Imaging Studies longitudinal dataset available on Kaggle is used for experiments. The dataset is balanced using synthetic minority oversampling technique. Experiments are done on both imbalanced and balanced datasets. Decision tree obtained 73.38% accuracy, XGB obtained 83.88% accuracy and RF obtained a maximum of 87.84% accuracy on the imbalanced dataset. Decision tree obtained 83.15% accuracy, XGB obtained 91.05% accuracy and RF obtained maximum 95.03% accuracy on the balanced dataset. A maximum accuracy of 95.03% is achieved with SMOTE-RF.

KEYWORDS

biomedical electronics, computer vision, feature extraction, medical signal processing

1 | INTRODUCTION

The most prevalent neurodegenerative disorder is Alzheimer's disease (AD). Its symptoms are firstly mild, but with time symptoms increase. Ten to 20 years before symptoms appear, the brain begins to change in the early stages of this disease. It gradually impairs thinking skills and damages memories. A group of symptoms linked to cognitive impairment make up dementia. Memory, thinking, reasoning, and the capacity to carry out daily duties are all impacted by dementia. The most typical cause of dementia is AD. Over 70% of dementia patients come from low-income nations. Dementia patients face difficulty in managing their emotions. Mostly, old age persons are affected by this disease. The person suffering from this illness could have anxiety or memory problems, such as forgetting familiar names and places. The person's close friends and family have noticed that they have trouble remembering their names. A doctor can identify a patient's memory and attention issues by performing a thorough medical interview [1].

Alzheimer's disease symptoms persist and worsen with time. This development impairs a person's capacity for efficient communication, environmental adaptation, and finally even the execution of simple movements. It gets harder for them to verbally express their pain or suffering. They frequently need significant support for daily tasks due to the continuing decrease in memory and cognitive abilities. At this point in the disease, Alzheimer's patients could encounter the following difficulties:

1. Everyday activities and personal care require round-the-clock assistance.
2. They lose awareness of their surroundings and recent events.
3. With time, their physical capabilities, such as sitting, walking, and swallowing, may change.
4. Interacting with others is getting harder and harder.
5. An increase in the prevalence of infections, particularly pneumonia.

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Although this disease has no treatment, its symptoms can be reduced with timely detection [2]. There are therapeutic options that can slow disease development using an early and precise diagnosis. The impact of machine learning (ML) techniques on the healthcare industry has grown significantly recently [3]. The pressure to create new methods for early identification and intervention of AD has increased due to the lack of a permanent treatment.

1.1 | Scenario

Alzheimer's disease develops gradually, starting with mild symptoms that get worse over time. A prompt and correct diagnosis is essential for enabling actions and support that may enhance the quality of life. This study aims to improve the accuracy and responsiveness of AD prediction. Early detection and intervention are crucial to improve patient outcomes and lower the societal and financial burden due to the ageing population and lack of a recognised treatment for the illness. Machine learning presents a promising approach for locating subtle patterns that can help in early diagnosis due to its ability to analyse complex data [4]. By giving medical practitioners early detection tools, this research has the potential to change clinical practice. We can significantly contribute to the ethical and social imperative of improving the lives of millions of people impacted by this disease by utilising ML for prediction.

1.2 | Motivations

This research is motivated by the pressing global health challenge posed by AD. There are two main motivations. First, the need for rapid diagnosis and treatment of Alzheimer's is made more urgent by the ageing world population. Second, there is an urgent need for automated technologies that can support medical professionals in their diagnostic work in the current diagnostic landscape.

1.3 | Proposed solution

In this research, a methodology for AD prediction called SMOTE-RF is proposed. Authors have carefully analysed the performance of three well-known ML algorithms: decision tree (DT), extreme gradient boosting (XGB), and random forest (RF) in their effort to create a reliable predictive model. The well-known Open Access Series of Imaging Studies (OASIS) dataset, which is available on Kaggle, is utilised to carry out experiments. Predictive model performance can be greatly impacted by class imbalance, a typical problem in medical data. Synthetic minority oversampling technique (SMOTE), a method that rebalances the dataset by creating synthetic samples for the minority class is used to overcome this issue. By ensuring that models are not biased towards the majority class, this stage enables more precise predictions.

1.4 | Main contributions

The Main Contributions of this research are as follows:

1. ML algorithms namely DT, XGB, and RF are used for model building to predict AD.
2. Experiments are performed in two ways, first on the original dataset and then on class balanced dataset.
3. As the dataset is highly imbalanced, so the class imbalance problem is overcome by the SMOTE technique.

Various existing systems of predicting Alzheimer's are discussed in section 2. The methodology used in this paper is explained in section 3. Results of experiments performed are demonstrated in section 4. Work is concluded in section 5.

2 | RELATED WORKS

Antor et al. [2] used Support Vector Machine (SVM), DT, RF, and logistic regression algorithms for the early detection of Alzheimer's. Eight attributes of OASIS dataset were used for prediction. Missing values in all features were deleted except socioeconomic status (SES). Missing values in this feature were filled using median. A ratio of 80:20 was used for training and testing data. Random forest suffered from the problem of overfitting. SVM provided the best accuracy at 92%. Kavitha et al. The authors in ref. [3] performed prediction using DT, SVM, RF, voting and gradient-boosting classifiers. The missing data of SES was filled using a median method. Random forest obtained the highest accuracy of 86.92%.

Leong and Abdullah [5] performed feature selection to select significant attributes from the OASIS-I cross-sectional dataset. There were 5.63% missing values in the dataset, and these values were removed from the dataset. Random forest provided the best accuracy of 94.39%. Alickovic and Subasi [6] used histograms for the extraction of features and RF for classification. Histogram transformed images of brain into feature vectors. After that the classification was done attaining 85.77% accuracy on the Alzheimer's Disease Neuroimaging Initiative (ADNI) dataset. Neelaveni [7] presented an AD prediction methodology by using psychological parameters. Two ML classifiers SVM and DT were applied for classification and best 85% accuracy was obtained by an SVM classifier. Shahbaj et al. [8] used K-Nearest Neighbours (KNN), DT, Naïve Bayes (NB), rule induction and linear model for predicting Alzheimer's. Algorithms were applied on the ADNI dataset. The highest accuracy of 88.24% was achieved with the linear model.

Velazquez et al. [9] considered nine features from the ADNI dataset for the prediction of Alzheimer's. Imbalanced classes were balanced using oversampling. After oversampling, prediction was done using RF attaining 93.6% accuracy. Bashir et al. [10] developed a deep neural network-based system for predicting Alzheimer's. The Open Access Series of Imaging Studies dataset was used for experiments, and 92.39% accuracy was obtained. Vashishtha et al. [11] performed feature selection

using wrapper methods. Experiments were performed with and without feature selection methods. Random forest, DT, SVM, XGB classifiers were used. Classifiers provided increased accuracy with feature selection.

Martinez-Murcia et al. [12] have presented the deep learning-based auto encoders for AD prediction. The features from MRI pictures had been extracted to describe a person's cognitive symptoms. The distribution of the collected features is then examined using classification analysis, and the effect of each coordinate of the auto encoder manifold on the brain is estimated.

According to studies [13], 30% of instances of AD can be delayed by the early detection and cure of modifiable risk factors [14]. The Lifestyle for Brain Health index is one method suggested by the Innovative Midlife Intervention for Alzheimer's Deterrence project [15, 16]. The three primary types of dementia intervention [17, 18] are cognitive training, increased physical activity and hypertension treatment. Alzheimer's disease is the most prevalent variety of diseases (AD). Vascular Alzheimer's, the other most prevalent form, is followed by Alzheimer's with Lewy bodies. Other forms of AD are linked to alcohol misuse, infections, and brain trauma.

In their work, Tatiq and Barber [19] hypothesised that AD could be avoided by focusing on vascular risk factors because these two types frequently coexist in the brain and share some vascular risk variables. Williams et al. [20] used four alternative models—DT, SVM, NN, and NB—to produce predictions of cognitive performance based on demographic and neuropsychological data. The accuracy of NB was the highest in this situation because average values were used to fill in the gaps left by the missing values.

Menagadevi et al. [21] presented an automated Alzheimer's detection model using MRI datasets. Two datasets were used for experiments: one is ADNI, and the other one is from Kaggle repository. Image preprocessing was done by curvelet thresholding and then image enhancement was done by octagon histogram equalisation. The classification was performed by SVM, KNN and extreme learning machine. The results were validated by the k-fold validation method with values of k 3, 5 and 10. The SVM classifier provided the best accuracy of 98.21% on the ADNI dataset and 99.77% on the Kaggle dataset.

Prasath et al. [22] proposed an AD detection model based on deep learning architectures. The MRI images dataset was taken from Kaggle repository. During the preprocessing stage, the images were resized and then image enhancement was done by fusion methods. The long-term potentiation features were extracted from the processed fused images and then fed into classifiers. The best 99.5% accuracy was obtained by the proposed Alzheimer's detection system.

Shukla et al. [23] presented an AD detection and diagnosis methodology using ML algorithms. The dataset set was gathered from the ADNI website and then images were converted from 4D to 2D format. Then clipping, histogram equalisation and grayscale conversion were done during the preprocessing stage. The feature extraction and selection were performed by the principal component analysis method. The dataset was split

into 7:3 for train and test, and classification was performed by Convolution Neural Networks, XGBoost and RF classifiers. The best accuracy of 97.57% was achieved by the proposed method.

3 | MATERIALS AND METHODS

3.1 | Dataset

The Open Access Series of Imaging Studies dataset available on Kaggle is used in experiments [24]. This dataset is a popular collection of clinical and neuroimaging data that is largely targeted towards AD research. Here is a detailed explanation of each feature:

1. **MRI ID:** This is an identification code assigned to each MRI session. It uniquely identifies each imaging session or scan.
2. **Subject ID:** A unique identification code is assigned to each subject in the dataset. This code allows researchers to associate multiple MRI sessions with the same individual.
3. **Age:** The age of the subject at the time of the MRI session. Age can be an important factor in AD research, as the risk of developing the disease often increases with age.
4. **M/F:** This feature indicates the sex of the subject. It typically uses 'M' for male and 'F' for female.
5. **Hand:** This feature represents the subject's significant hand of use, which can be either right-handed (R), left-handed (L), or ambi-dextrous (both hands) (B).
6. **EDUC:** The number of years of education received by the subject. Education level can be relevant when studying the cognitive function and AD.
7. **SES (Socio-Economic Status):** SES is a 5-level categorisation of the subject's socio-economic status, which reflects their societal class:
 - Level 1: Lower class
 - Level 2: Lower-middle class
 - Level 3: Middle class
 - Level 4: Middle-upper class
 - Level 5: Upper class
8. **Mini-Mental State Examination (MMSE) Score:** MMSE is a cognitive assessment score that measures various cognitive abilities. Scores range from 0 to 30, where lower scores indicate a higher likelihood of dementia, and higher scores suggest better cognitive health.
9. **Clinical Diagnosis Rating (CDR):** CDR is a clinical rating that categorises the subject's cognitive status based on assessments, including MMSE and MRI scans. It has four levels:
 - CDR = 0: Cognitively normal
 - CDR = 0.5: Very mild dementia
 - CDR = 1: Mild dementia
 - CDR = 2: Moderate dementia
10. **Estimated Total Intracranial Volume (eTIV):** This represents the estimated total volume of the brain, often measured in unspecified units.

11. **Normalised Whole Brain Volume (nWBV):** nWBV is a measure of the nWBV. It represents the proportion of the brain's total volume relative to the eTIV.
12. **Atlas Scaling Factor (ASF):** ASF is the determinant of an affine transformation matrix applied to brain MRI data points.
13. **Delay:** The interval, measured in days, between the previous and the current MRI session. This can be relevant for tracking changes over time.
14. **Visit:** This feature indicates the ordinal number of the visit to the testing facility for the MRI session. It helps track the sequence of visits for each subject.
15. **Group:** The dementia group to which the subject belongs. It is categorised into three levels:
 - Dementia: Indicates significant dementia.
 - Converted: Denotes subjects who transitioned to a significant dementia state after the initial assessment.
 - Non-Demented: Represents subjects who do not have dementia.

There are three output classes: demented, non-demented, and converted represented by 0, 1, and 2. The distribution of output classes is shown in Figure 1.

Data preprocessing has been done by removing unnecessary attributes: Subject ID, MRI ID, and visit. The correlation between features is shown in Figure 2 and the histogram of features is shown in Figure 3. Analysis of Figure 1 indicates that the distribution of output classes is not even. The uneven distribution of classes can adversely affect the performance of the models when trained. Therefore, SMOTE was applied to make the even distribution of classes.

3.2 | Methodology

The proposed methodology for predicting Alzheimer's is shown in Figure 4. The collection of the dataset pertinent to AD is the first step in the methodology. Data preprocessing is then carried out to make sure the dataset is appropriate for additional analysis. During data preprocessing, irrelevant attributes are eliminated, and the dataset is balanced. Subject ID, MRI ID, and visit attributes are removed from the dataset. The number of output classes was not uniform. Therefore, balancing is done using SMOTE. Synthetic minority over-sampling technique creates synthetic data points for minority classes, resulting in a more accurate representation of all output classes.

The methodology's next step involves applying classification algorithms to data that has been preprocessed to produce predictions about AD. Three classifiers DT, XGB, and RF are used for classification. Decision tree is known for its ability to capture complex decision boundaries in the data. XGBoost is an ensemble learning method that has the ability to handle complex datasets. Random forest is good in lowering over-fitting and improving the model's predictive capability. In order to produce predictions about the output class, which in this

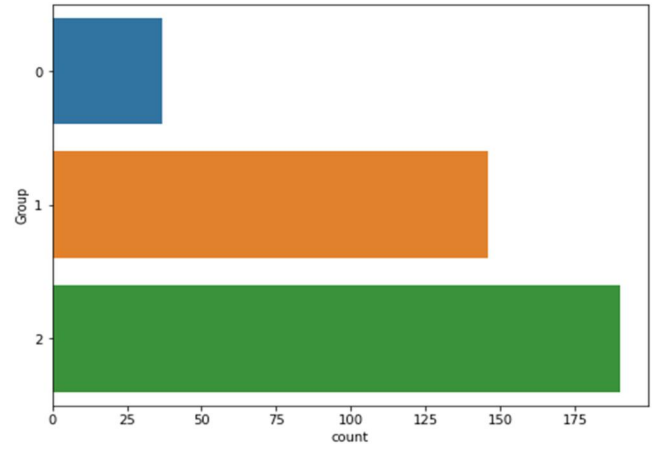


FIGURE 1 Distribution of output classes.

case refers to the likelihood of AD, trained classifiers are applied to the preprocessed data.

The system's early prediction capability, which is essential for prompt intervention and treatment, is one of its strong points. It lessens the chance of human mistakes in diagnosis and prognosis. This might result in more trustworthy outcomes. The model has some limitations, including the possibility of False positives (FP) (forecasting Alzheimer's when it is not there) and false negatives (failing to detect Alzheimer's when it is present). Patients may experience unneeded stress in the case of FP, and diagnoses may be missed in the case of false negatives. The knowledge of healthcare practitioners is complemented, not replaced, by the proposed model. To understand the model's predictions and decide on patient care, doctors and specialists are required.

The performance of classifiers is calculated in terms of accuracy, sensitivity, specificity, precision, and F-measure.

Accuracy: Calculates the percentage of predictions made by the system that are accurate.

$$\text{Accuracy} = \left(\frac{\text{TP} + \text{FP}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \right) * 100$$

Sensitivity: Measures the system's accuracy rate for positive predictions.

$$\text{Sensitivity} = \left(\frac{\text{TP}}{\text{FN} + \text{TP}} \right) * 100$$

Specificity: Measures the system's accuracy rate for negative predictions.

$$\text{Specificity} = \left(\frac{\text{TN}}{\text{FP} + \text{TN}} \right) * 100$$

Precision: Calculates the percentage of relevant findings for the system produced.

FIGURE 2 Correlation between features.

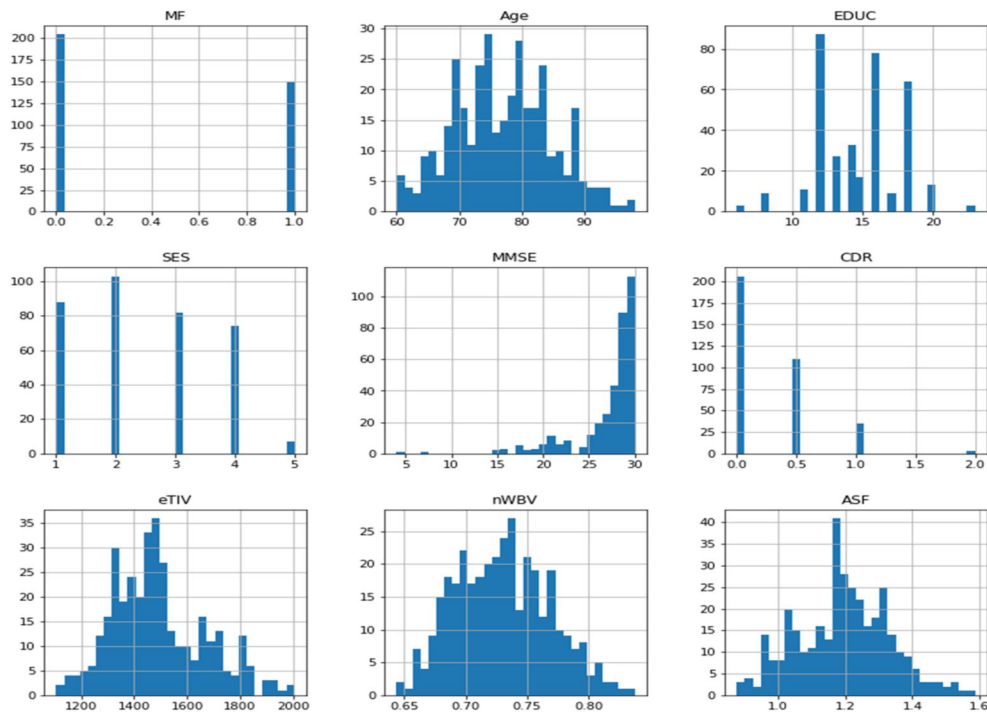
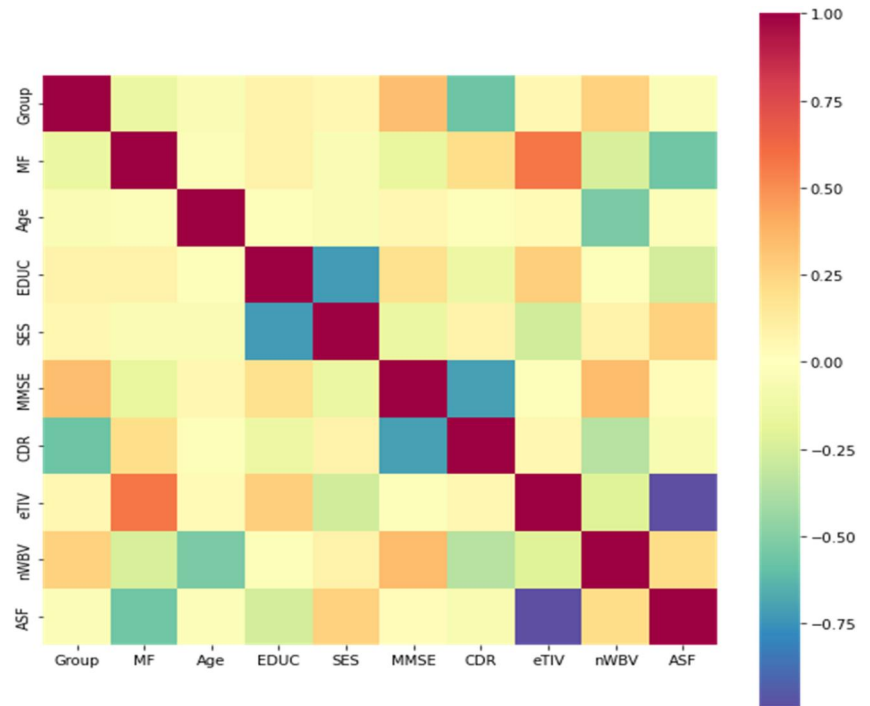


FIGURE 3 Histogram of features.

$$\text{Precision} = \left(\frac{\text{TP}}{\text{FP} + \text{TP}} \right) * 100$$

F-Measure: To calculate F-Measure, the harmonic means of sensitivity and precision is used.

$$\text{F-Measure} = 2 * \frac{\text{Sensitivity} * \text{Precision}}{\text{Sensitivity} + \text{Precision}}$$

The number of correctly detected negatively classified cases is denoted by the acronym TN, which stands for true negatives. The term “true positives,” or “TP,” refers to the

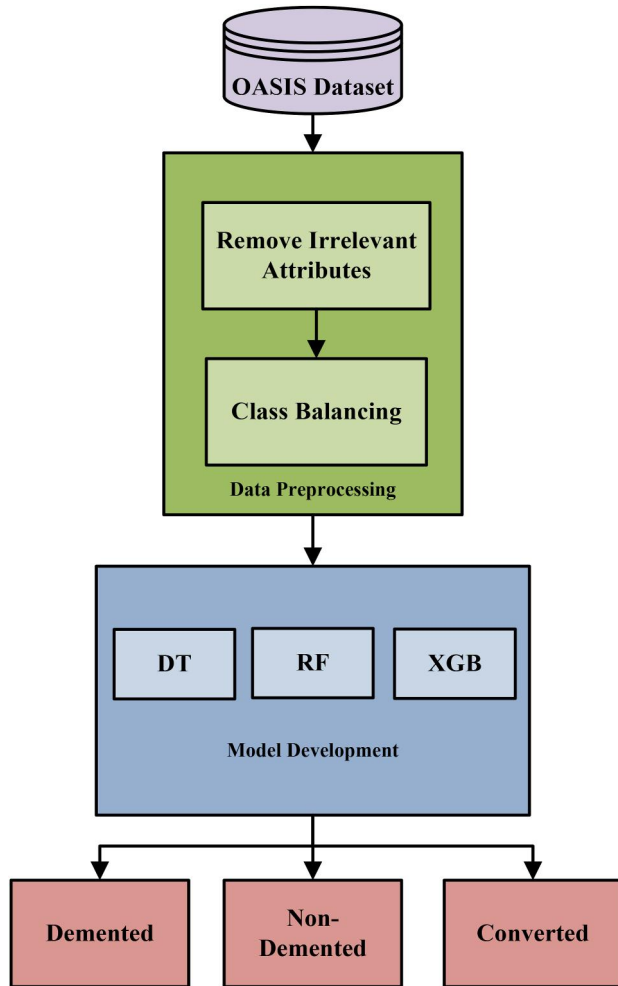


FIGURE 4 Proposed methodology for predicting Alzheimer's disease (AD).

quantity of correctly identified positive cases. False negatives, or FN, are positive cases that were inadvertently classified as negative. False positives, denoted by the initials FP, are the number of negative cases that were incorrectly classified as positive [25, 26].

3.3 | Classifiers

Three Classifiers used in the proposed methodology are described in this section. Decision tree has a root node, internal nodes, and leaf nodes. Nodes are connected through branches. At every internal node, the attribute's value is tested and the test's result is on the branch. Leaf nodes contain the class labels that represent the outcome. A root node is the parent of all other nodes. In DT, features are represented by internal nodes, and rules are represented by branches. Data is arranged in a tree manner, processing a particular outcome at each leaf. It is very easy to gather the data and come up with some insightful insights because

decision trees imitate the thinking of humans [27, 28]. The structure of DT is demonstrated in Figure 5.

One of the most effective ways to create gradient-boosted decision trees is XGBoost. It is specifically developed to optimise memory utilisation. Building successive sub-trees from an original tree to reduce the errors of the preceding one is the core concept behind boosting. The new sub-trees will so update the older ones. In this manner, the new sub-trees will update the earlier residuals to lessen the cost function's error. Random forest is a set of trees trained using samples obtained from a random resampling of the training set [29, 30]. Bootstrap samples are those produced by randomly resampling the training set. A collection of bootstraps that exclude records from the original dataset is used as the test set once the tree has been formed. Each tree casts a vote for one class, and the forest calculates which class will receive the majority of votes to categorise the input data [31, 32]. Random forest is shown in Figure 6.

4 | RESULTS AND DISCUSSION

There are two sets of experiments. Experiments are done using three classifiers on the original dataset and balanced dataset. The performance of classifiers on the original imbalanced dataset is shown in Table 1. Decision tree obtained 73.38% accuracy, XGB obtained 83.88% accuracy and RF obtained a maximum of 87.84% accuracy. Decision tree obtained 84.67% precision, XGB obtained 91.33% precision and RF obtained 95.17% precision. Decision tree obtained 84.67% specificity, XGB obtained 87.21% specificity and RF obtained 87.68% specificity. Very low sensitivity was obtained by all classifiers in the imbalanced dataset. Decision tree obtained 19.04% sensitivity whereas XGB and RF obtained only 15% sensitivity. Results in Table 1 indicate that sensitivity is quite low in all classifiers. This was due to an imbalanced dataset. Therefore, balancing is further done to improve performance. The performance of classifiers after balancing is shown in Table 2.

Balancing increased performance of all parameters, there was an effective improvement in sensitivity. Decision tree obtained 91.71% sensitivity, XGB obtained 93.85% sensitivity and RF obtained a maximum of 95.26% sensitivity. In addition to sensitivity, other performance parameters of classifiers also improved. Decision tree obtained 83.15% accuracy, XGB obtained 91.05% accuracy and RF obtained a maximum of 94.03% accuracy. Decision tree obtained 85.63% precision, XGB obtained 93.68% precision and RF obtained 95.26% precision. Decision tree obtained 85.63% specificity, XGB obtained 94.17% specificity and RF obtained 94.27% specificity. Decision tree obtained 88.46% F-measure, XGB obtained 93.93% F-measure and RF obtained 94.76% F-measure. Classifiers' performance comparison before and after balancing is shown in Figures 7–11. The analysis of these figures indicates that balancing has significantly improved the performance of all the classifiers, and RF has provided the highest value of all the performance parameters.

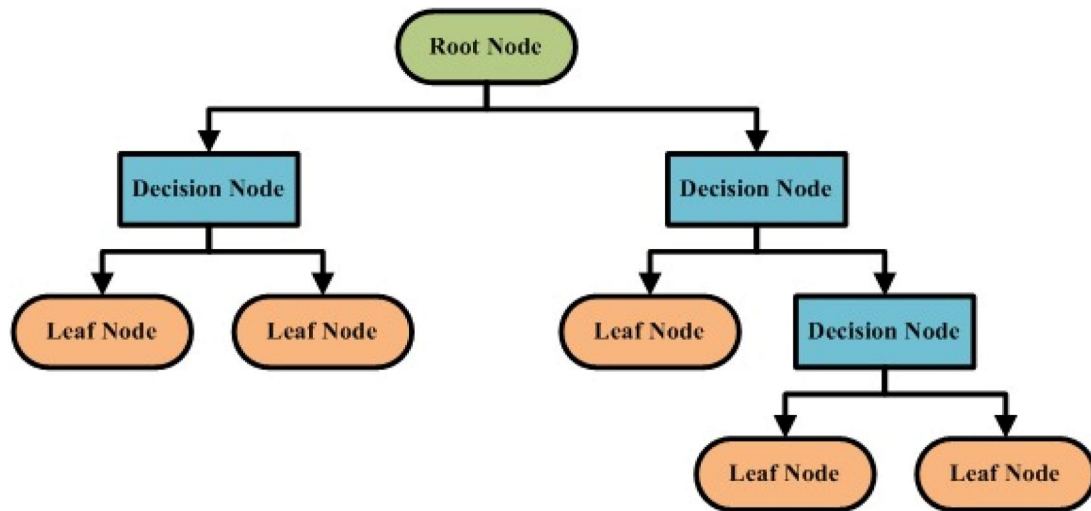


FIGURE 5 Structure of decision tree (DT).

FIGURE 6 Random forest (RF).

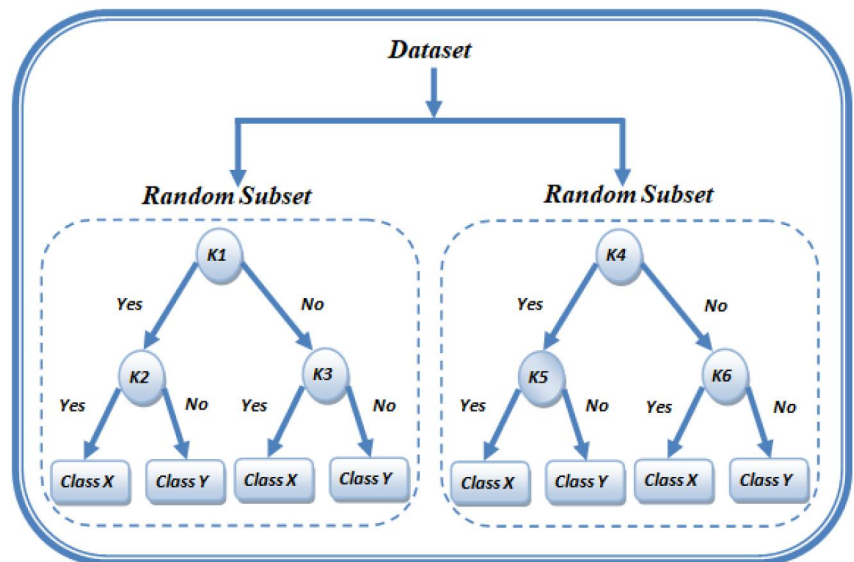


TABLE 1 Performance of classifiers on the imbalanced dataset.

Classifier	Accuracy	Precision	Sensitivity	Specificity	F-measure
DT	73.38	84.67	19.04	84.67	85.36
RF	87.84	95.17	15.00	87.68	91.32
XGB	83.88	91.33	15.00	87.21	89.23

TABLE 2 Performance of models on the balanced dataset.

Classifier	Accuracy	Precision	Sensitivity	Specificity	F-measure
DT	83.15	85.63	91.71	85.63	88.46
RF	94.03	95.26	93.95	94.27	94.76
XGB	91.05	93.68	93.85	94.17	93.93

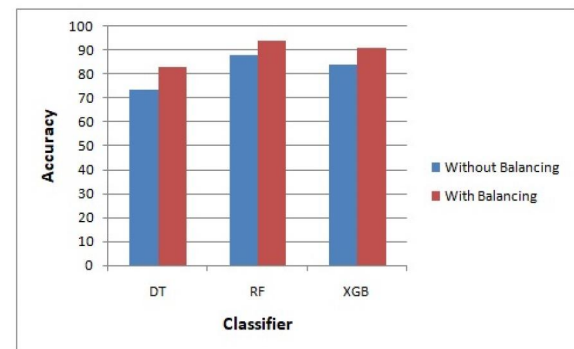


FIGURE 7 Classifiers' accuracy comparison.

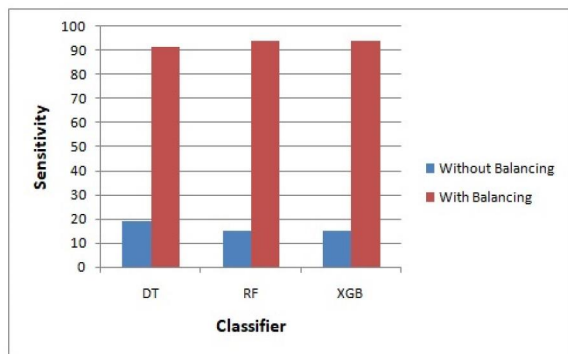


FIGURE 8 Classifiers' sensitivity comparison.

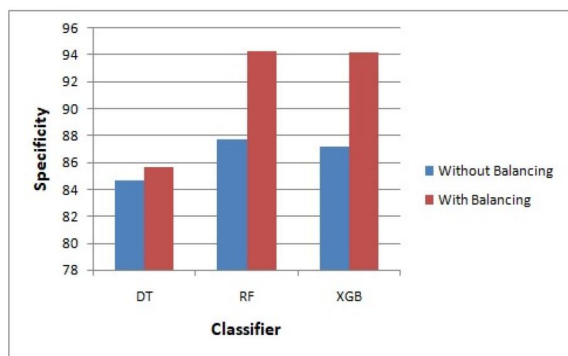


FIGURE 9 Classifiers' specificity comparison.

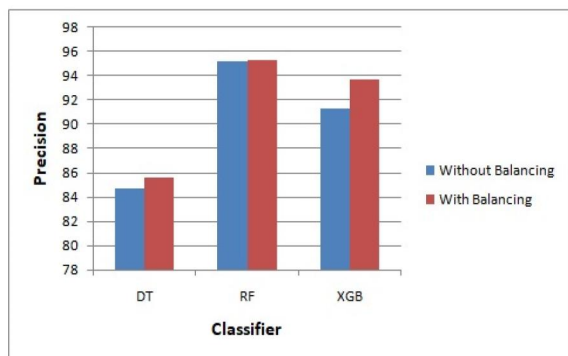


FIGURE 10 Classifiers' precision comparison.

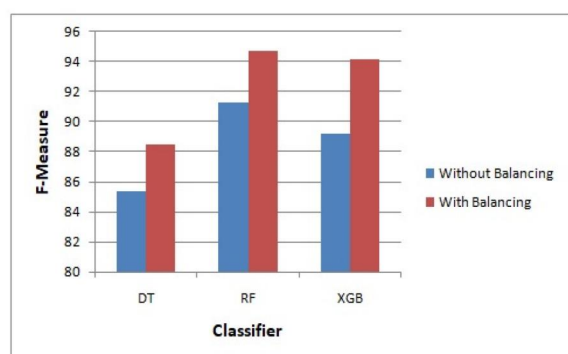


FIGURE 11 Classifiers' F-measure comparison.

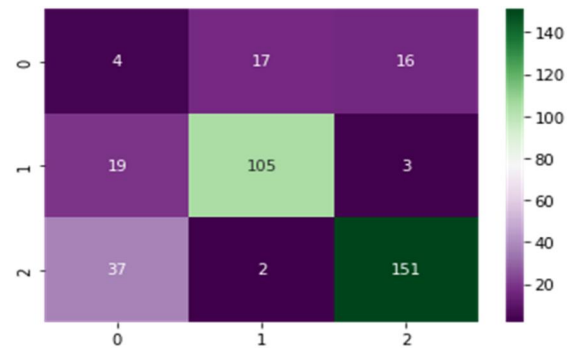


FIGURE 12 Decision tree (DT) confusion matrix without balancing.

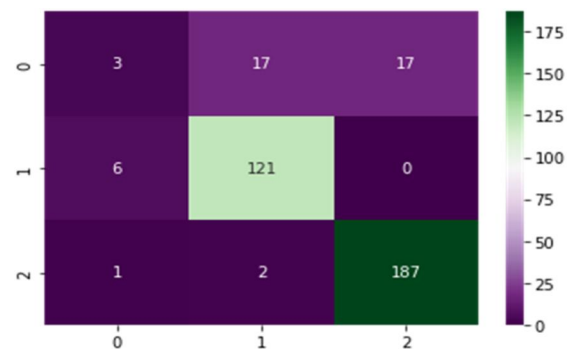


FIGURE 13 Random forest (RF) confusion matrix without balancing.

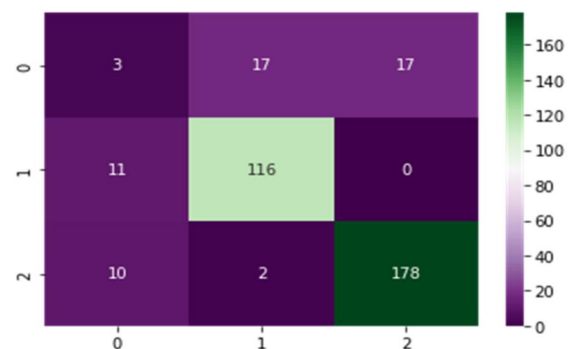


FIGURE 14 Extreme gradient boosting (XGB) confusion matrix without balancing.

The classifier's confusion matrix is also evaluated to determine performance. Confusion matrices show counts of expected and observed values. The anticipated and actual classification done by the classifier is displayed in a confusion matrix of size $n \times n$, where n is the total number of classes. By examining the diagonal values for determining the number of accurate classifications, one might assess the model's accuracy by visualising the confusion matrix. The classifier's confusion matrix before and after balancing is shown in Figures 12–17. There was an increase in correct predictions after balancing.

Experimental analysis indicated that the RF along with SMOTE provided the best performance. Comparison of work is also done with recent existing methodologies in Table 3.

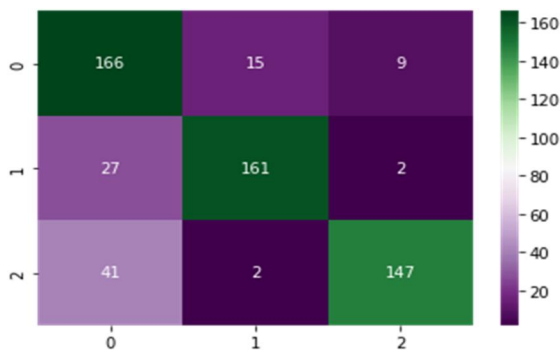


FIGURE 15 Decision tree (DT) confusion matrix with balancing.

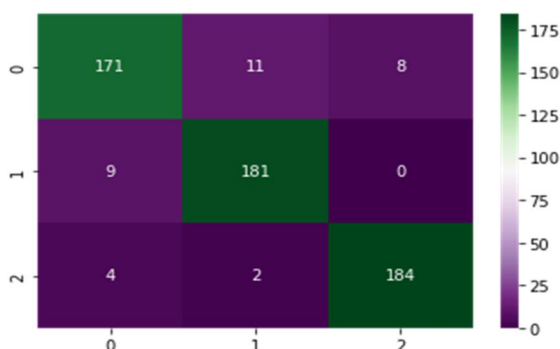


FIGURE 16 Random forest (RF) confusion matrix with balancing.

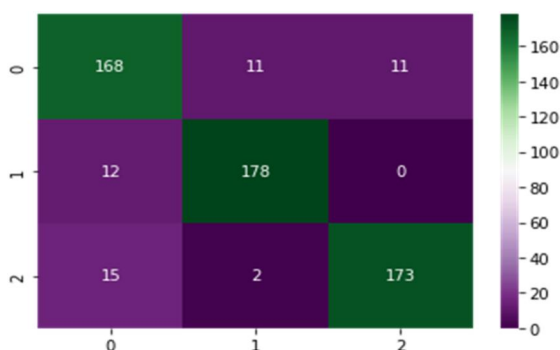


FIGURE 17 Extreme gradient boosting (XGB) confusion matrix with balancing.

Existing research used different ML methods to categorise AD using the OASIS dataset. SMOTE-RF provided better results than recent works in literature. The maximum accuracy was reached by the SMOTE-RF, which merged SMOTE and RF, with a 94.03% accuracy, demonstrating its better performance. Improvement in performance with SMOTE-RF as compared to existing methods is shown in Figure 18.

This study's limitation is its reliance on pre-existing neuroimaging datasets such as OASIS, which by nature have little demographic variety, potential biases, and difficulties generalising to real-world clinical scenarios. The model's capacity to adjust to varied demographics and data sources may be hampered by its reliance on a single dataset for both training

TABLE 3 Comparison with existing methodologies.

Authors	Year	Dataset	Methodology	Accuracy
Antor et al. [1]	2021	OASIS	SVM	92%
Basheer et al. [9]	2021	OASIS	DNN	92.39%
Kavitha et al. [2]	2022	OASIS	RF	86.92%
Proposed methodology	-	OASIS	SMOTE-RF	94.03%

Abbreviation: SVM, Support Vector Machine.

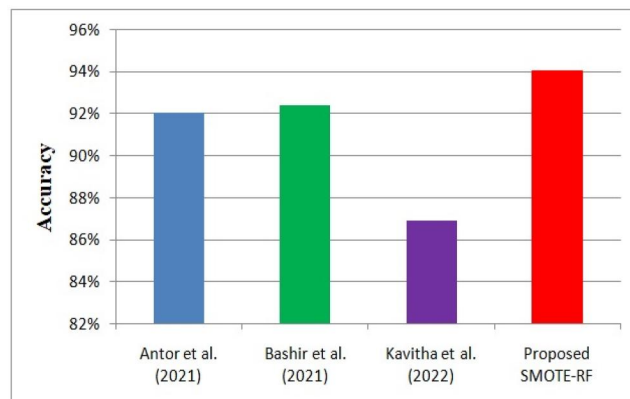


FIGURE 18 Improvement in performance with SMOTE-RF.

and evaluation. Despite the model's promise in research environments, integrating it seamlessly into clinical practice is a difficult and time-consuming procedure that requires regulatory permissions and validation in clinical settings, which could delay its practical deployment.

5 | CONCLUSION & FUTURE WORK

Early diagnosis of Alzheimer's can help in slowing the disease's development. In this paper, DT, RF, and XGB ML algorithms are used to predict Alzheimer's. Experiments are performed without balancing and with balancing. Balancing was done using SMOTE. The Open Access Series of Imaging Studies dataset available on Kaggle is used for experiments. Without balancing, DT, XGB, and RF obtained 73.38%, 83.88%, and 87.84% accuracy, respectively. With balancing, DT provided 83.15% accuracy, XGB provided 91.05% accuracy and RF provided 94.03% accuracy.

In future, authors will develop a system with improved accuracy using ensemble methods. Work can also be extended to perform prediction using brain images.

AUTHOR CONTRIBUTIONS

Pooja Rani: Investigation; methodology; software. **Rohit Lamba:** Data curation; resources; writing – original draft. **Ravi Sachdeva:** formal analysis; validation; visualisation. **Karan Kumar:** Conceptualisation; validation; writing – review & editing. **Celestine Iwendi:** Formal analysis; software; visualisation.

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CONFLICT OF INTEREST STATEMENT

The authors declare that they have no known competing financial interests.

DATA AVAILABILITY STATEMENT

The Open Access Series of Imaging Studies dataset is used that is available on Kaggle. https://www.kaggle.com/datasets/jboysen/mri-and-alzheimers?select=oasis_longitudinal.csv (accessed Aug. 10, 2022).

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